KAZI NIMAT HASAN

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RESEARCH INTERESTS

Bioinformatics & Machine Learning | Medical Image Analysis | Biomaterials and Bio-nanomaterials | Hydrogel-based Drug Delivery

EDUCATION

Khulna University of Engineering & Technology *CGPA: 3.80/4.00 (Ranked 6th in overall merit position)*

Bachelor of Science in Biomedical Engineering *March, 2022- August, 2026*

- **UG THESIS:** Integrated Transcriptomic Analysis for Host-Directed Target Discovery in Measles Virus Infection
- Integrated public bulk and single-cell RNA-seq datasets to characterize persistent host transcriptional responses during measles virus infection
- Performed differential gene expression, functional enrichment, and protein–protein interaction network analysis to distinguish responses in infected vs. bystander cells and identify candidate host-directed therapeutic targets

Higher Secondary Certificate (HSC) | GPA: 5:00/5.00 *2018- 2020*

Notre Dame College (English Version)

Secondary School Certificate (SSC) | GPA: 5:00/5.00 *2016-2018*

Dhaka Residential Model College (English Version)

PUBLICATIONS

- M.A. Sayed, M. A. Kadir, M. S. Islam, A. Rahman, M. N. Hossain, J. Park, M. S. I. Wadud, R. T. Asha, **K. N. Hasan**, and H. Mehdi, "Enhancing lung cancer diagnosis with region-of-interest guided classification, segmentation, and PET-like feature extraction from CT scans," *Scientific Reports*, under revision. Contribution: annotation of ~1,000 CT images for dataset preparation; formal analysis; manuscript review and editing
- M. A. Sayed, M. A. Kadir, N. Mostakim, M. J. Hossain, M. S. Alam, **K. N. Hasan**, and S. Hossain, "Advanced image analysis for liver tumor detection and visualization in CT images using automated segmentation," in preparation for resubmission. Contribution: manual annotation of ~1,000 CT images; manuscript formatting and revision.
- Thesis-based manuscript (in preparation).

PROJECTS

AI-Driven Glucose Monitoring System for Telemedicine Application | *BME 4232*| May 2026

- Built a low-cost glucose estimator using phototransistor sensing and a Random Forest model in Python

Assistive System for Paralyzed Patients Using Motion & Wireless Communication| *BME 3200*| Oct-Feb 25

- Built a gesture-based aid for paralyzed patients with a custom foil-graphite flex sensor and alerts

IoT-Based Baby Monitoring System & Smart Cradle | *ECE 3216*| Oct-Jan 2025

- Built a low-cost IoT smart cradle (Arduino, NodeMCU) monitoring crying and environment

Design and Analysis of A Pacemaker Electrode using COMSOL Multiphysics | *BME 3114, KUET* | Mar-Jun 2024

- Simulated a pacemaker electrode in COMSOL, comparing materials and diameters for best performance

Closed-Loop Controlled Anesthesia with BIS Monitoring| *BME 3112, KUET* | Feb-May 2024

- Designed a closed-loop anesthesia control concept using EEG-derived BIS feedback

IoT-Based Patient Health Monitoring System| *EEE Day* | Nov 2023

- Built a wearable band tracking heart rate, temperature, and BP, streaming to a custom Bluetooth app

TRAINING AND CERTIFICATIONS

- Industrial Attachment in Biomedical Engineering | National Electro-Medical Equipment Maintenance Workshop & Training Center and Technical College of Allied Health | Specialized government hospitals | 2026
- Machine Learning with Python (80 hrs) | CCC, KUET (EDGE Project, Bangladesh Computer Council) | Oct – Dec 2024
- Workshop: Research Design & Thesis Writing using LaTeX | Advanced Bio-Engineering Club, KUET

TECHNICAL SKILLS

- **Data Science & Bioinformatics:** Python; machine learning (Random Forest); bulk and single-cell RNA-seq analysis; differential expression; functional enrichment; protein–protein interaction networks.
- **Hardware & Embedded**: Arduino, NodeMCU (ESP8266), MPU6050, RF/Bluetooth modules, IoT prototyping,
- **Simulation & Analysis:** COMSOL Multiphysics; MATLAB; Ansys.
- **Laboratory & Characterization:** Hydrogel preparation; freeze-thaw crosslinking; freeze-drying; swelling; water retention; gel fraction; degradation

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Vice President, Biomedical Engineering Association, KUET *Sep 2025 – Sep 2026*

- Highest-ranking student representative in the department; organized the departmental study tour, the 2K19 farewell, new-student orientation, a seminar on the Code of Conduct for Prevention of Sexual Harassment, and BME Indoor 4.0

Vice President, TRY (Non-profit Voluntary Organization of KUETians) *Aug 2025 – Aug 2026*

- Led a winter clothing drive for 180 people in Abhaynagar, Jashore; an Iftar program for 123 people across three madrasas in Khulna; and Zakat distribution and fundraising campaigns

Head of Administration, KUET Adventure Club *Sep 2025 – Jul 2026*

- Organized the national event "Tales in Frames" with 150+ participants; earlier Assistant Secretary of Administration (Mar 2024 – Aug 2025) and Executive, Press & Publication (Mar 2023 – Feb 2024)

Junior Organizing Secretary, Advanced Bio-Engineering Club of KUET (ABC KUET) *Apr 2023 – Mar 2024*
Class Representative, Department of Biomedical Engineering, KUET *Mar 2022 – Jan 2023*

HONORS & AWARDS

- Dean's Award, GPA 3.86/4.00 (3rd year), KUET, Sep 2026
- Bronze Award for Voluntary Blood Donation | Voluntary Blood Donation Society of KUET (DREAM) | Nov 2025
- ALOHA Mental Arithmetic: **International Grand Champion**, Group J (Malaysia, 2013)
- ALOHA Mental Arithmetic: **2 times National Champion**, Level 3 (2012, 2013)
- BME Indoor 4.0: Champion, Table Tennis (Singles) and Cards; Runner-up, Table Tennis (Doubles) | BME Association, KUET | Dec 2024

REFERENCES

- **Dr. Mostafa Zaman Chowdhury,** Professor, Department of Electrical and Electronic Engineering, KUET — mzceee@yahoo.com
- Dr. G.M. Mahmudur Rahman, Associate Professor, Department of Biomedical Engineering, KUET — mahmud@bme.kuet.ac.bd
- **Jeba Fairooz Rahman**, Assistant Professor, Department of Biomedical Engineering, KUET — jeba@bme.kuet.ac.bd